15.1 INTRODUCTION

In the previous chapter, we discussed the interference pattern produced by division of a wavefront; for example, light coming out of a pinhole was allowed to fall on two holes, and spherical waves emanating from these two holes produced the interference pattern. In this chapter, we will consider the formation of interference pattern by division of amplitude; for example, if a plane wave falls on a thin film then the wave reflected from the upper surface interferes with the wave reflected from the lower surface. Such studies have many practical applications and also explain phenomena like the formation of beautiful colors produced by a soap film illuminated by white light.

15.2 INTERFERENCE BY A PLANE PARALLEL FILM WHEN ILLUMINATED BY A PLANE WAVE LO 1

If a plane wave is incident normally on a thin* film of uniform thickness d (see Fig. 15.1) then the waves reflected from the upper surface interfere with the waves reflected from the lower surface. In this section, we will study this interference pattern. In order to observe the interference pattern without obstructing the incident beam, we use a partially reflecting plate G as shown in Fig. 15.1. Such an arrangement also enables us to eliminate the direct beam from reaching the photographic plate P (or the eye). The plane wave may be produced by placing an illuminated pinhole at the focal point of a corrected lens; alternatively, it may just be a beam coming out of a laser.

Let the solid and the dashed lines in Fig. 15.2 represent the positions of the crests** (at any particular instant of time) corresponding to the waves reflected from the upper and lower surfaces of the film, respectively.*** Clearly, the wave reflected from the lower surface of the film traverses an additional optical path of $2nd$, where n represents the refractive index of the material of the film. Further, if the film is placed in air, then the wave reflected from the upper surface of the film will undergo a sudden change in phase of π (see Sec. 14.12) and as such the conditions for destructive or constructive interference will be given by

$$2nd = m\lambda \quad \text{destructive interference} \qquad \textbf{(15.1a)}$$

$$= \left(m + \frac{1}{2}\right)\lambda \quad \text{constructive interference} \qquad \textbf{(15.1b)}$$

where m = 0, 1, 2,…and λ represents the free space wavelength.

Thus, if we place a photographic plate at P (see Fig. 15.1), then the plate will receive uniform illumination; it will be dark when $2nd = m\lambda$ and bright when $2nd = \left(m + \frac{1}{2}\right)\lambda$; m = 0, 1, 2,… Instead of placing the photographic plate, if we try

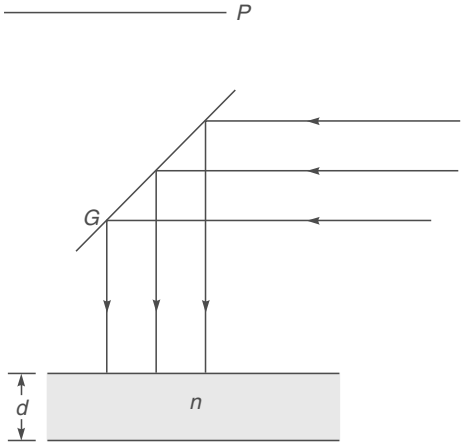


Fig. 15.1 The normal incidence of a parallel beam of light on a thin film of refractive index n and thickness d. G denotes a partially reflecting plate and P represents a photographic plate.

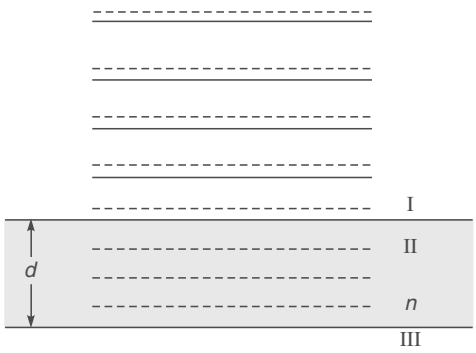


Fig. 15.2 The solid and the dashed lines represent the crests of the waves reflected from the upper surface and from the lower surface of the thin film. Notice that the distance between the consecutive crests inside the film is less than the corresponding distance in medium I.

* Why the film should be thin is explained in Sec. 15.7.

** Notice that the distance between consecutive crests in the film is less than the corresponding distance in air. This is because of the fact that the effective wavelength in a medium of refractive index n is λ/n.

*** In general, the wave reflected from the lower surface of the film will suffer multiple reflections. The effect of such multiple reflections is neglected (see Chapter 16).

to view the film (from the top) with naked eye, then the film will appear to be uniformly illuminated.

It may be noted that the amplitudes of the waves reflected from the upper and lower surfaces will, in general, be slightly different; and as such the interference will not be completely destructive. However, with appropriate choice of the refractive indices of media II and III, the two amplitudes can be made very nearly equal (see Example 15.1).

For an air film between two glass plates (see Fig. 15.3) no phase change will occur on reflection at the glass-air interface, but a phase change of π will occur on reflection at the air-glass interface and the conditions for maxima and minima will remain the same. On the other hand, if the I medium is crown glass (n = 1.52), the II medium is an oil of refractive index 1.60 and the III medium is flint glass (n = 1.66) then a phase change of π will occur at both the reflections and the conditions for maxima and minima would be

$$2nd = \left(m + \frac{1}{2}\right)\lambda \qquad \text{minima} \qquad \textbf{(15.2a)}$$

$$= m\lambda \qquad \text{maxima} \qquad \textbf{(15.2b)}$$

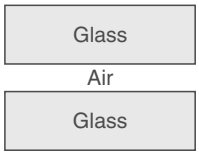


Fig. 15.3 Thin film of air formed between two glass plates.

In general, whenever the refractive index of the II medium lies in between the refractive indices of the I and the III media, then the conditions of maxima and minima would be given by Eqs. (15.2a) and (15.2b).

We next consider the oblique incidence of the plane wave on the thin film (see Fig. 15.4). Once again, the wave reflected from the upper surface of the film interferes with the wave reflected from the lower surface of the film. The latter traverses an additional optical path Δ, which is given by (see Fig. 15.5):

$$\Delta = n_2(BD + DF) - n_1 BC \qquad \textbf{(15.3)}$$

where C is the foot of the perpendicular from the point F on BG. We will show in the next section that

$$\Delta = 2n_2 d \cos\theta' \qquad \textbf{(15.4)}$$

where θ' is the angle of refraction.

For a film placed in air, a phase change of π will occur when reflection takes place at the point B and as such, the conditions of destructive and constructive interference would be given by

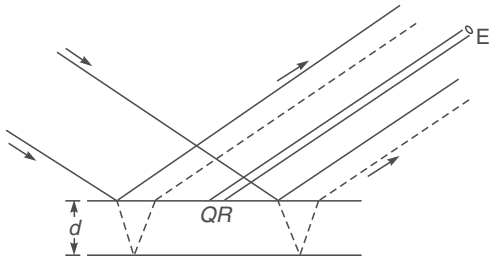


Fig. 15.4 The oblique incidence of a plane wave on a thin film. The solid and dashed lines denote the boundary of the wave reflected from the upper surface and from the lower surface of the film. The eye E receives the light reflected from the region QR.

$$\Delta = 2n_2 d \cos\theta' = m\lambda \qquad \text{minima} \qquad \textbf{(15.5a)}$$

$$= \left(m + \frac{1}{2}\right)\lambda \qquad \text{maxima} \qquad \textbf{(15.5b)}$$

If we place a photographic plate at P (see Fig. 15.5) it will receive uniform illumination; if we try to view the film with naked eye (at the position E — see Fig. 15.4) then only light rays reflected from a small position QR of the film will reach the eye. The image formed at the retina will be dark or bright depending on the value of Δ (see Eq. 15.5).

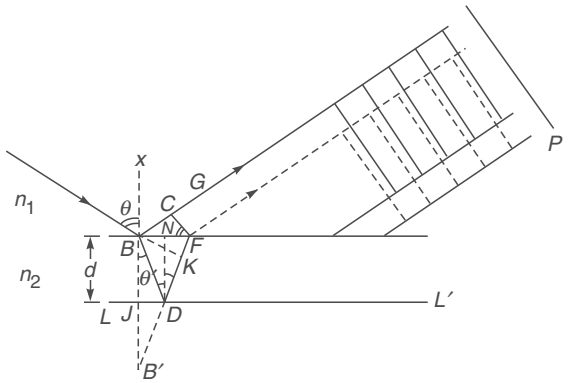


Fig. 15.5 Calculation of the optical path difference between the waves reflected from the upper surface of the film and from the lower surface of the film. The solid and the dashed lines represent the corresponding positions of the crests. P denotes a photographic plate.

15.3 THE COSINE LAW LO 2

In this section, we will show that the wave reflected from the lower surface of the film traverses an additional optical path which is given by the following expression:

$$\Delta\,[= n_2(BD + DF) - n_1 BC] = 2n_2 d \cos\theta' \qquad \textbf{(15.6)}$$

Let θ and θ' denote the angles of incidence and refraction respectively. We drop a perpendicular BJ from the point B on the lower surface LL' and extend BJ and FD to the point B' where they meet (see Fig. 15.5). Clearly,

$$\angle JBD = \angle BDN = \angle NDF = \theta'$$

where N is the foot of the perpendicular drawn from the point D on BF. Now,

$$\angle BDJ = \frac{\pi}{2} - \theta'$$

and $$\angle B'DJ = \pi - \left[\left(\frac{\pi}{2} - \theta'\right) + \theta' + \theta'\right] = \frac{\pi}{2} - \theta'$$

Thus, $BD = BD'$ and $BJ = JB' = d$

or $BD + DF = B'D + DF = B'F$

Hence, $$\Delta = n_2 B'F - n_1 BC \qquad \textbf{(15.7)}$$

Now, $\angle CFB = \angle CBX = \theta$

$$BC = BF \sin\theta = \frac{KF}{\sin\theta'} \sin\theta = \frac{n_2}{n_1} KF \qquad \textbf{(15.8)}$$

where K is the foot of the perpendicular from B on $B'F$. Substituting the above expression for BC in Eq. (15.7), we get

$$\Delta = n_2 B'F - n_2 KF = n_2 B'K$$

or $$\Delta = 2n_2 d \cos\theta' \qquad \textbf{(15.9)}$$

which is known as the **cosine law**.

15.4 NON-REFLECTING FILMS

LO 3

One of the important applications of the thin film interference phenomenon discussed in Sec. 15.2 lies in reducing the reflectivity of lens surfaces; this we plan to discuss in this section. However, for a quantitative understanding of the phenomenon, we will have to assume that when a light beam (propagating in a medium of refractive index n_1) is incident normally on a dielectric of refractive index n_2 then the amplitudes of the reflected and the transmitted beams are related to that of the incident beam through the following relations* [see Fig. 15.6(a)]:

$$a_r = \frac{n_1 - n_2}{n_1 + n_2} a_i \qquad \textbf{(15.10a)}$$

$$a_t = \frac{2n_1}{n_1 + n_2} a_i \qquad \textbf{(15.10b)}$$

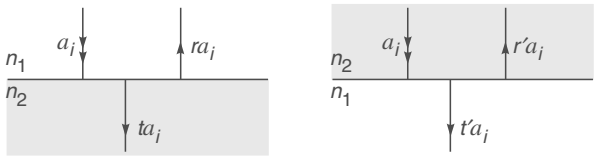


Fig. 15.6 (a) If a plane wave of amplitude a_i, propagating in a medium of refractive index n_1, is incident normally on a medium of refractive index n_2, then the amplitudes of the reflected and the transmitted beams are a_r and a_t, respectively. Similarly, (b) corresponds to the case when the beam (propagating in a medium of refractive index n_2) is incident on a medium of refractive index n_1.

where a_i, a_r and a_t are the amplitudes of the incident beam, reflected beam and the transmitted beam respectively. Notice that when $n_2 > n_1$, a_r is negative showing that when a reflection occurs at a denser medium a phase change of π occurs. The amplitude reflection and transmission coefficients r and t are, therefore, given by [see Sec. 24.2]:

$$r = \frac{n_1 - n_2}{n_1 + n_2} \qquad \textbf{(15.11a)}$$

$$t = \frac{2n_1}{n_1 + n_2} \qquad \textbf{(15.11b)}$$

If r' and t' are the reflection and transmission coefficients where light propagating in a medium of refractive index n_2 is incident on a medium of refractive index n_1 [see Fig. 15.6(b)], then

$$r' = \frac{n_2 - n_1}{n_2 + n_1} = -r \qquad \textbf{(15.12)}$$

$$t' = \frac{2n_2}{n_1 + n_2} \qquad \textbf{(15.13)}$$

and

$$1 - tt' = 1 - \frac{4n_1 n_2}{(n_1 + n_2)^2} = \left(\frac{n_1 - n_2}{n_1 + n_2}\right)^2 = r^2 \qquad \textbf{(15.14)}$$

Equations (15.13) and (15.14) represent the Stokes' relations (see Sec. 14.12).

We will now discuss the application of the thin film interference phenomenon in reducing the reflectivity of lens surfaces. We all know that in many optical instruments (like a telescope) there are many interfaces and the loss of intensity due to reflections can be severe. For example, for near normal incidence,** the reflectivity of crown glass surface in air is

* These relations can be derived from electromagnetic theory; see Eqs. (24.67)–(24.72) (with $\theta_1 = \theta_2 = 0$).

** In all what follows in this section, we will assume near normal incidence.

$$\left(\frac{n-1}{n+1}\right)^2 = \left(\frac{1.5-1}{1.5+1}\right)^2 \simeq 0.04,$$

i.e., 4% of the incident light is reflected. For a dense flint glass $n \simeq 1.67$ and about 6% of light is reflected. Thus, if we have a large number of surfaces, the losses at the interfaces can be considerable. In order to reduce these losses, lens surfaces are often coated with a $\lambda/4n$ thick 'non-reflecting film'; the refractive index of the film being less than that of the lens. For example, glass ($n = 1.5$) may be coated with an MgF_2 film (see Fig. 15.7) and the film thickness d should be such that*

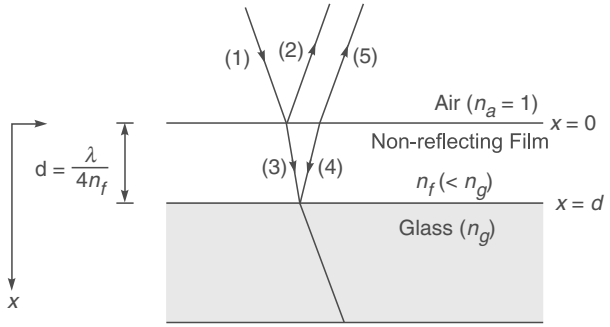


Fig. 15.7 If a film (having a thickness of $\lambda/4n_f$ and having refractive index less than that of the glass) is coated on the glass, then waves reflected from the upper surface of the film destructively interfere with the waves reflected from the lower surface of the film. Such a film is known as a non-reflecting film.

$$2n_f d = \frac{1}{2}\lambda$$

or
$$d = \frac{\lambda}{4n_f} \qquad \textbf{(15.15)}$$

where we have assumed near normal incidence [i.e., $\cos\theta' \cong 1$; see Eq. (15.9)] and n_f represents the refractive index of the film; for MgF_2, $n_f = 1.38$. Thus, if we assume λ to be 5.0×10^{-5} cm (which roughly corresponds to the center of the visible spectrum), we will have

$$d = \frac{5.0 \times 10^{-5}\ \text{cm}}{4 \times 1.38} \approx 0.9 \times 10^{-5}\ \text{cm}$$

Figure 15.8 shows a comparison between a grasses lens without anti-reflective coating (top) and a lens with anti-reflective coating (bottom). Note the reflection of the photographer in the top lens and the tinted reflection in the bottom. We would like to emphasize the following points:

(a) Let n_a, n_f and n_g be the refractive indices of air, non-reflecting film and glass, respectively. If a is the amplitude of the incident wave then the amplitudes of the reflected and refracted waves (the corresponding rays shown as (2) and (3) in Fig. 15.7) would be

$$-\frac{n_f - n_a}{n_f + n_a}\, a \text{ and } \frac{2n_a}{n_f + n_a}\, a$$

respectively (we have assumed near normal incidence). The amplitudes of the waves corresponding to rays (4) and (5) would be

$$-\frac{2n_a}{n_f + n_a}\frac{n_g - n_f}{n_g + n_f}\, a$$

and
$$-\frac{2n_a}{n_f + n_a}\frac{n_g - n_f}{n_g + n_f}\frac{2n_f}{n_f + n_a}\, a$$

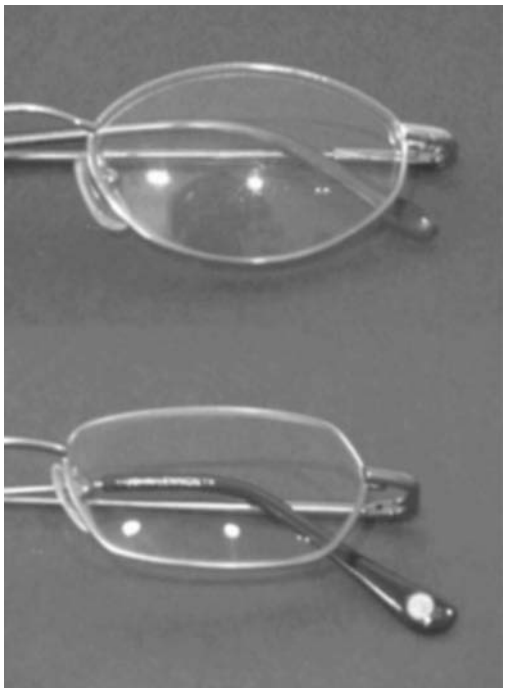


Fig. 15.8 Comparison between a glasses lens without anti-reflective coating (top) and a lens with anti-reflective coating (bottom). Note the reflection of the photographer in the top lens and the tinted reflection in the bottom. [*Figure adapted from http://en.wikipedia.org/wiki/Optical_coating (Photograph taken by Justin Lebar); a color photograph appears as Fig. 18 in the prelim pages–used with permission of Mr. J. Lebar*].

* Since the refractive index of the non-reflecting film is greater than that of air and less than that of the glass, abrupt phase change of π occurs at both the reflections. Consequently, when $2nd \cos\theta' = m\lambda$ there would be constructive interference and when $2nd \cos\theta' = (m + \frac{1}{2})\lambda$ there would be destructive interference.

respectively. Now, for complete destructive interference, the waves corresponding to rays (2) and (5) should have the same amplitude, i.e.,

$$-\frac{n_f - n_a}{n_f + n_a} a = -\frac{2n_a}{n_f + n_a}\frac{n_g - n_f}{n_g + n_f}\frac{2n_f}{n_f + n_a} a \quad \textbf{(15.16)}$$

$$\text{or} \quad \frac{n_f - n_a}{n_f + n_a} = \frac{n_g - n_f}{n_g + n_f} \quad \textbf{(15.17)}$$

where we have used the fact that $\frac{4n_a n_f}{(n_f + n_a)^2}$ is very nearly equal to unity; for $n_a = 1$ and $n_f = 1.4$,

$$\frac{4n_a n_f}{(n_f + n_a)^2} \approx 0.97$$

On simplification we obtain

$$n_f = \sqrt{n_a n_g} \quad \textbf{(15.18)}$$

If the first medium is air then $n_a = 1$ and with $n_g = 1.66$ (dense flint glass) n_f should be 1.29 and when $n_g = 1.5$ (light crown glass) n_f should be 1.22. We note that the refractive indices of magnesium fluoride and cryolite are 1.38 and 1.36, respectively. Now for a $\frac{\lambda}{4n}$ thick film, the reflectivity will be about

$$\left[\frac{n_f - n_a}{n_f + n_a} - \frac{n_g - n_f}{n_g + n_f}\right]^2 \quad \textbf{(15.19)}$$

Thus, for $n_a = 1$, $n_f = 1.38$ and $n_g = 1.5$ the reflectivity will be about 1.3%. In the absence of the film, the reflectivity would have been about 4%. The reduction of reflectivity is much more pronounced for the dense flint glass. This technique of reducing the reflectivity is known as **blooming**.

(b) The film is non-reflecting only for a particular value of λ; in Eq. (15.15) λ was assumed to be 5000 Å. For a polychromatic light, the film's non-reflecting property will be falling off when λ is greater or less than the above value. However, the effect is not serious. For example, for the MgF_2 film on crown glass at 5000 Å, the reflectivity rises by about 0.5% as one goes either to the red or the violet end of the visible spectrum. In Sec. 15.4.2, we will discuss why we should use a $\frac{\lambda}{4n}$ thick film and not $\frac{3\lambda}{4n}$ or $\frac{5\lambda}{4n}$ thick film, although the latter will also give destructive interference for the chosen wavelength.

(c) As in the case of Young's double slit experiment there is no loss of energy; there is merely a redistribution of energy. The energy appears mostly in the transmitted beam.

15.4.1 Mathematical Expressions for the Reflected Waves

It may be worthwhile to carry out a bit of mathematical analysis for the anti-reflecting film shown in Fig. 15.7. We assume $n_g > n_f > n_a$ and that the x-axis is pointing downwards with $x = 0$ at the upper surface of the film. The displacement associated with the incident wave (propagating in the $+x$ direction) is given by

$$y_1 = a \cos(\omega t - k_a x); \quad k_a = \frac{\omega}{c} n_a \quad \textbf{(15.20)}$$

Thus at $x = 0$, $y_1 = a \cos \omega t$. The reflected wave (shown as 2) would therefore be

$$y_2 = -a|r_1| \cos(\omega t + k_a x); \quad \textbf{(15.21)}$$

where

$$|r_1| = \left|\frac{n_f - n_a}{n_f + n_a}\right| \quad \textbf{(15.22)}$$

is a positive quantity. The minus sign in Eq. (15.21) represents the sudden phase change of π at $x = 0$. The transmitted wave (shown as 3) would be given by

$$y_3 = at_1 \cos(\omega t - k_f x); \quad k_f = \frac{\omega}{c} n_f \quad \textbf{(15.23)}$$

where

$$t_1 = \frac{2n_a}{n_f + n_a} \quad \textbf{(15.24)}$$

Thus the displacement at $x = d$ [associated with wave (3)] is

$$y_3 = at_1 \cos(\omega t - k_f d) \quad \textbf{(15.25)}$$

Therefore, the wave reflected from the lower surface [wave (4), which would be propagating in the negative x-direction] is given by

$$y_4 = -at_1|r_2| \cos[\omega t + k_f(x - 2d)]$$

$$|r_2| = \left|\frac{n_g - n_f}{n_g + n_f}\right| \quad \textbf{(15.26)}$$

where the phase factor is adjusted such that at $x = +d$ we obtain the phase given by Eq. (15.25). The wave (5) would therefore be given by

$$y_5 = -at_1|r_2|t_2 \cos[\omega t + k_a x - 2k_f d] \quad \textbf{(15.27)}$$

Assuming the amplitudes of y_2 and y_5 to be approximately the same, destructive interference (between y_2 and y_5) would occur if

$$2k_f d = \pi, 3\pi, \ldots \quad \textbf{(15.28)}$$

or

$$d = \frac{\lambda_f}{4}, \frac{3\lambda_f}{4}, \frac{5\lambda_f}{4}, \ldots \quad \lambda_f = \frac{\lambda}{n_f} \quad \textbf{(15.29)}$$

15.4.2 Rigorous Expressions for Reflectivity

In the above section, we have considered two-beam interference and have neglected multiple reflections at the lower and upper surfaces. The effect of multiple reflections will be discussed in Sec. 16.2; however, such an effect is automatically taken into account when we solve Maxwell's equations incorporating the appropriate boundary conditions. In Sec. 24.4, we will carry out such an analysis and will show that the reflectivity (at normal incidence) of a dielectric film of the type shown in Fig. 15.7 is given by [see Sec. 24.10 and Eq. (24.94)]:*

$$R = \frac{r_1^2 + r_2^2 + 2r_1 r_2 \cos 2\delta}{1 + r_1^2 r_2^2 + 2r_1 r_2 \cos 2\delta} \qquad \textbf{(15.30)}$$

where

$$r_1 = \frac{n_a - n_f}{n_a + n_f} \quad \text{and} \quad r_2 = \frac{n_f - n_g}{n_f + n_g} \qquad \textbf{(15.31)}$$

represent the Fresnel reflection coefficients at the first and second interface, respectively, and

$$\delta = \frac{2\pi}{\lambda} n_f d \qquad \textbf{(15.32)}$$

d being the thickness of the film and, as before, λ represents the free space wavelength. Elementary differentiation shows us that $dR/d\delta = 0$ when $\sin 2\delta = 0$. Indeed for $r_1 r_2 > 0$,

$$\cos 2\delta = -1 \text{ (minima)} \qquad \textbf{(15.33)}$$

represents the condition for minimum reflectivity and when this condition is satisfied, the reflectivity is given by

$$R = \left(\frac{r_1 - r_2}{1 - r_1 r_2}\right)^2 = \left(\frac{n_a n_g - n_f^2}{n_a n_g + n_f^2}\right)^2 \qquad \textbf{(15.34)}$$

where we have used Eq. (15.18). Thus the film is non-reflecting when

$$n_f = \sqrt{n_a n_g}$$

consistent with Eq. (15.18). Now, the condition $\cos 2\delta = -1$ implies

$$2\delta = \frac{4\pi}{\lambda} n_f d = (2m+1)\pi;\ m = 0, 1, 2, \ldots \qquad \textbf{(15.35)}$$

or

$$d = \frac{\lambda}{4n_f}, \frac{3\lambda}{4n_f}, \frac{5\lambda}{4n_f}, \ldots \qquad \textbf{(15.36)}$$

In Fig. 15.9, we have plotted the reflectivity as a function of δ for

$$n_a = 1 \qquad n_g = 1.5 \qquad \textbf{(15.37)}$$

and

$$n_f = \sqrt{n_a n_g} \simeq 1.225$$

As expected, R is maximum ($\approx 4\%$) when $\delta = 0, \pi, 2\pi, \ldots$ and the film is anti-reflecting ($R = 0$) when $\delta = \pi/2, 3\pi/2, \ldots$ implying $d = \lambda/4n_f, 3\lambda/4n_f, \ldots$ As an example, let us suppose that we wish to make the film antireflecting at $\lambda = 6000$ Å; then from Eq. (15.26), the thickness of the film could be

$$1224.5 \text{ Å} \quad \text{or} \quad 3673.5 \text{ Å} \quad \text{or} \quad 6122.5 \text{ Å}, \ldots$$

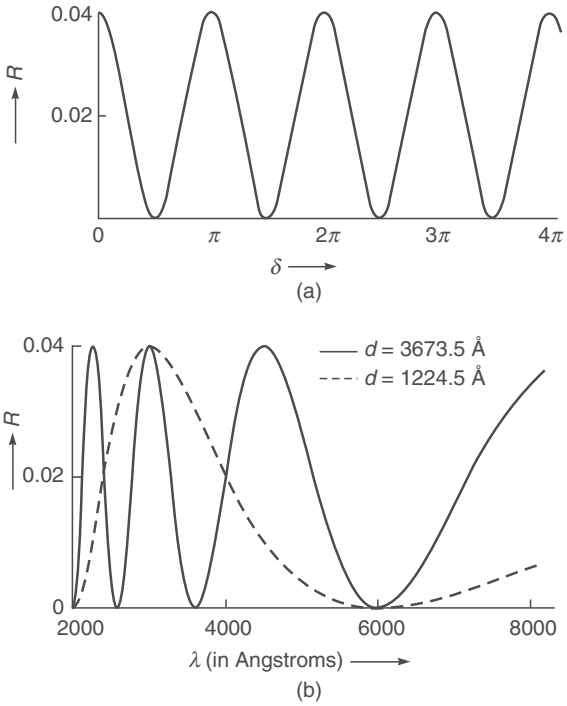


Fig. 15.9 (a) Variation of the reflectivity of a film as a function of δ ($= 2\pi n_f d/\lambda$) for $n_a = 1$, $n_g = 1.5$ and $n_f = \sqrt{n_a n_g} \simeq 1.225$. Notice that the reflectivity is zero for $\delta = \pi/2, 3\pi/2, 5\pi/2, \ldots$ (b) Wavelength variation of the reflectivity for a film of thickness 1224.5 Å (dashed curve) and of thickness 3673.5 Å (solid curve) with $n_a = 1$, $n_g = 1.5$ and $n_f = \sqrt{n_a n_g} \approx 1.225$. Notice that both films are non-reflecting at 6000 Å.

* Equation (15.30) is actually valid even for oblique incidence with r_1, r_2 and δ defined appropriately (see Sec. 24.10).